INCH-POUND

MIL-DTL-32610
w/AMENDMENT 1
07 July 2021
SUPERSEDING
MIL-DTL-32610
27 October 2018

DETAIL SPECIFICATION

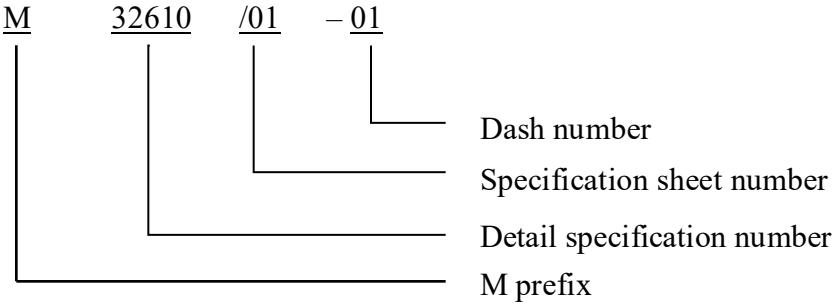
STRIPPER, MULTI-CONDUCTOR ELECTRICAL CABLE, HAND-OPERATED, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the design and test requirements for tools and accessories to strip aerospace vehicle twisted multi-conductor cables with twisted multiple shielded conductors. The stripper is primarily used to strip twisted multi-conductor cable intended for Electrical Wiring Interconnect System (EWIS) within aircraft applications.

1.2 Part or Identifying Number (PIN). The PINs to be used for the strippers acquired in accordance to this specification are created as follows:



Comments, suggestions, or questions on this document should be addressed to Commanding Officer, Naval Air Warfare Center Aircraft Division Lakehurst, System Standardization and PHS&T Branch, Code BL32600, Mail Stop 120-3, Route 547, Joint Base MDL, NJ 08733-5100 or emailed to LKE-NAVAIRDocComments@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at https://assist.dla.mil.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATION

MIL-DTL-32610/1	-	Stripper, Multi-Conductor Electrical Cable, (for Conductor Diameters 0.090 Through 0.180 Inch)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-202-103	-	Method 103, Humidity (Steady State)
MIL-STD-202-213	-	Method 213, Shock (Specified Pulse)

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

SAE INTERNATIONAL

SAE AIR1351	-	Manufacturers' Identification of Aerospace Electrical and Electronic Wiring Devices and Accessories

(Copies of this document are available online at http://www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual part requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

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3.2 First article. When specified (see 6.3.c), stripper samples shall be subjected to first article testing in accordance with 4.3.

3.3 Stripper. The stripper shall consist of the basic tool and stripping dies. The dies are removable and replaceable to accommodate different twisted pair cable configurations.

3.4 Materials. The materials used in the manufacturing of the stripper and accessories covered by this specification shall be capable of meeting all of the performance and test requirements of this specification and applicable specification sheet. The stripper, including all parts, shall be constructed and finished so that it is free of all defects that affect proper functioning in service.

3.4.1 Finish. All metal parts shall withstand the humidity test of 4.6.4.

3.5 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.6 Visual examination. The stripper shall have no burrs, sharp edges, nor sharp corners. The marking on the stripper shall be in accordance with the marking legibility requirements of 3.7.3.

3.6.1 Dimension. When inspected in accordance with 4.5.1, the dimensions of the stripper shall be in accordance with the applicable specification sheet.

3.7 Design and construction. The stripper shall be hand-operated and be capable of stripping the jacket or insulation of cables and wires specified in this specification and applicable specification sheets. Stripper and accessories shall be constructed to allow maintenance and repair required by normal operation and handling. The stripper shall consist of all components required to strip the applicable wires or cables. The stripper and all accessories shall be constructed to withstand stress, shock, vibration, and any other conditions incident to service use. The design and construction of the stripper shall be as specified herein and the specification sheet. Design shall conform to all performance requirements in 3.8.

3.7.1 Stripper user instructions. If specified in the contract (see 6.3.d), each stripper with accessories shall be provided with written user instructions. The stripper shall be capable of being assembled and disassembled in accordance with the user instructions without the use of special tools. The user instructions shall include the following information:

a. Military and manufacturer part numbers.

b. Pictorials. The instructions shall include pictorials of the stripper and accessories, which include the military part number and the manufacturer's part number. Pictorials shall be used wherever possible to illustrate the assembly, disassembly, maintenance, adjustments and operation of the stripper and all the accessories.

c. Safety. Instructions shall be provided so that the stripper and its accessories can be operated safely.

d. Operation. Instructions shall be provided in the selection and operation of the stripper and accessories. The instructions shall include application and specific wire and cable part numbers that can be stripped with the applicable conductor stripper die. Instructions shall also

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include stripper performance criteria and inspection method to ensure correct stripper operation and minimize cable/conductor damage (see 3.8.1).

e. Maintenance. Maintenance intervals and procedures shall be included in the stripper user instructions.

f. Replacement parts list. All replaceable parts designated in this specification and the specification sheet shall be included in the instructions. All parts that can be removed to assemble accessories shall be designated in the instructions by part number or a physical description sufficient to permit replacement (e.g., screw thread and length description, etc.).

g. Supplier. Instructions shall include address and telephone number(s) for the manufacturer of the stripper.

3.7.2 Weight (see 4.5.3). The weight of the stripper with all accessories attached shall be as specified in the specification sheet.

3.7.3 Marking (see 4.5.4). Markings including the PIN (see 1.2) shall be durable and withstand handling and usage of the stripper. Markings shall be clearly readable after all specified performance testing of this specification. Trademarks and logos when used for identification, rather than the manufacturer's name and address, shall be registered in SAE AIR1351.

3.7.4 Strip length. The minimum and maximum strip length cut by the cable stripper shall be specified in the specification sheet.

3.7.5 Cable diameter. The maximum cable diameter that the stripper can be stripped by the cable stripper manufactured in accordance with this specification shall be specified in the specification sheet.

3.7.6 Calibration. The stripper shall not require any calibration other than adjustment for the type of wire or cable being stripped. The stripper shall be adjusted prior to initial installation of dies for the wire or cable type being stripped. The stripper shall not require any additional adjustments or repair during the number of cycles specified in the life requirement.

3.7.7 Gages. Gages shall not be required for the calibration or adjustments of the stripper.

3.7.8 Wire or cables to be stripped by MIL-DTL-32610 wire or cable strippers. The type of wire or cable to be stripped by cable strippers manufactured in accordance with this specification shall be as specified in the specification sheet.

3.8 Performance.

3.8.1 Insulation stripping (see 4.6.1). The stripper shall be designed to follow the contour of the cable and shall have a controlled rotation that follows the cable contour and guides the cutting blades in a precise path along the length of the cable when stripping. The stripper shall have adjustable blades to allow the user to adjust the depth of the cutting blades. After adjustment in accordance with the user instructions (see 3.7.1), the blades shall score or cut the outer jacket of the cable without damaging the underlying shielding or multi-conductor insulation. For a two (2) conductor cable, the stripper shall produce two cuts longitudinally on the cable (see figure 1). For a three (3) conductor cable, the stripper shall produce a single cut longitudinally on the cable (see figure 2). The wire or cable conductor, or the shield may exhibit nicks or scrapes, but the base metal shall not be exposed. Broken or cut wire/cable conductor or shield strands are not acceptable.

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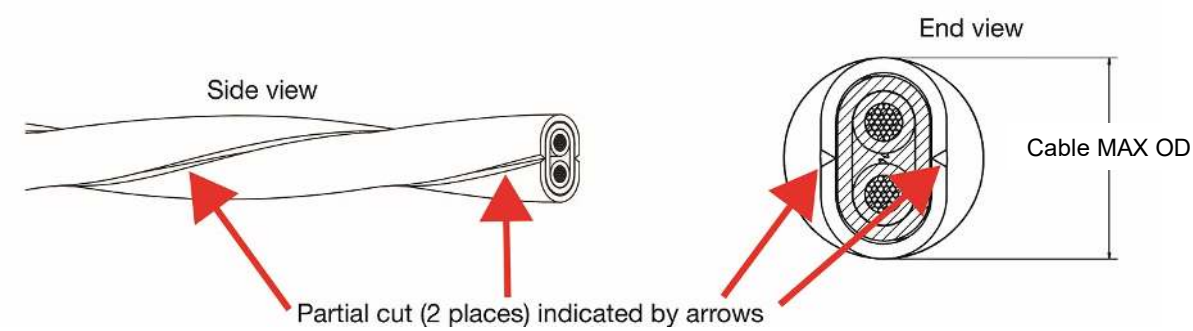


FIGURE 1. Two conductor cable longitudinal cut.

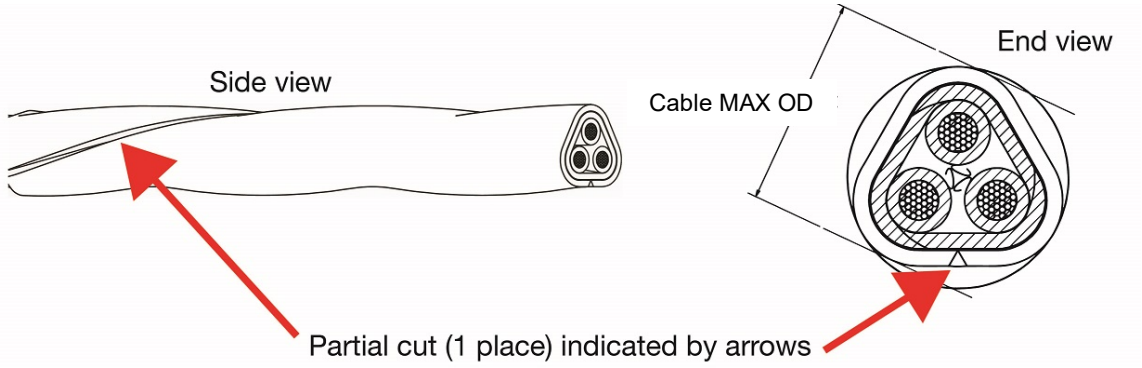


FIGURE 2. Three conductor cable longitudinal cut.

3.8.2 Stripping force. When tested in accordance with 4.6.2, the maximum strip force shall not exceed the stripping force specified in the specification sheet.

3.8.3 Shock. The stripper with required accessories shall meet the stripper insulation stripping-requirements of 3.7.3 and 3.8.1, after testing in accordance with 4.6.3.

3.8.4 Humidity. The stripper and attached accessories shall meet the marking and insulation stripping requirements of 3.8.1, after testing in accordance with 4.6.4.

3.8.5 Life. The stripper and attached accessories shall strip the wire or cable size and type specified in the specification sheet for the number of cycles specified in the specification sheet. After completion of the life test of 4.6.5, the stripping performance test of 4.6.1 shall be conducted and the stripper and dies shall meet the insulation stripping requirement of 3.8.1.

3.8.6 Workmanship (see 4.6.6). The stripper and dies, including all parts, shall be so constructed and finished that they shall be free of all defects that may affect proper functioning in service. The strippers shall be uniform in appearance without any mold flash. There shall be no sharp edges, burrs, or other defects that are potential hazards to the operator of the tool.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article test inspection (see 4.3).

b. Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified herein, all inspections shall be made at temperatures of 15 °C to 35 °C (59 °F to 95 °F) and at ambient air pressure and humidity. Whenever these conditions must be closely controlled in order to obtain reproducible results, for reference purposes, a temperature of 25° +0 °C, -2 °C (77 °F, +0 °F, -3.6 °F), relative humidity of 50 ±2 percent, and atmospheric pressure of 650 to 800 millimeters of mercury shall be used.

4.2.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy and quantity to permit performance of the required inspection shall be established and maintained by the contractor.

4.3 First article inspection. First article inspection shall be performed on three complete stripper assemblies. This inspection shall include examinations and tests specified in table I. The Original Equipment Manufacturer (OEM) (see 6.4) shall perform or witness the tests specified in table I.

TABLE I. First article and conformance inspections and tests.

Inspection	Requirement paragraph	Test paragraph
Visual and dimensional examination	3.6, 3.6.1	4.5.1, 4.5.2
Weight	3.7.2	4.5.3
Marking	3.7.3	4.5.4
Insulation stripping	3.8.1	4.6.1
Stripping force	3.8.2	4.6.2
Shock	3.8.3	4.6.3
Humidity 1/	3.8.4	4.6.4
Life 1/ 2/	3.8.5	4.6.5
Workmanship 2/	3.8.6	4.6.6

1/ Use one stripper for life testing and a second tool for humidity testing.
2/ Required only for initial first article testing.

4.3.1 First article inspection by similarity. When life test is completed on the required die set, all dies conforming to the dimensional and configuration requirements of the specification sheet shall be considered to have passed the life test provided that all die sets are manufactured using the same process, materials, and design as the die set tested.

4.4 Conformance inspection. Conformance inspections shall consist of the following tests.

4.4.1 Inspection of product for delivery. Inspection of product for delivery shall consist of conformance inspection test in accordance with 4.6.1 and 4.6.2.

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4.4.1.1 Inspection lot. An inspection lot shall consist of stripper assemblies under essentially the same manufacturing conditions and presented for inspection at one time.

4.4.1.2 Sampling plan. Statistical sampling shall be in accordance with table II. A randomly selected group(s) of samples is required. For acceptance of the lot, there shall be zero occurrences of defects.

TABLE II. Zero defect sampling plan.

Lot size	Minimum number of strippers to be tested
1 – 4	All
5 – 50	5
51 – 90	7
91 – 150	11
151 – 280	13
281 – 500	16
501 – 1200	19
1201 – 3200	23
3201 – 10000	29
10001 – 35000	35

4.5 Examinations.

4.5.1 Visual and dimensional examination (see 3.6 and 3.6.1). Strippers shall be examined to verify that the materials, design, construction, and physical dimensions and characteristics are in accordance with this specification and the specification sheet. Dimensions shall be measured with calibrated devices using standard laboratory methodology.

4.5.2 Examination of product. Strippers shall be examined to determine compliance with the requirements of this specification with respect to product identification, dimensions, and workmanship as specified in 3.6, 3.6.1, and 3.8.6.

4.5.3 Tool weight (see 3.7.2). The stripper shall be weighed on a calibrated scale with an accuracy of ±0.01 pound.

4.5.4 Marking examination (see 3.7.3). The stripper and accessories shall be examined for required marking and legibility with 1X magnification.

4.6 Test methods.

4.6.1 Insulation stripping performance (see 3.8.1).

4.6.1.1 Stripper adjustment. Prior to beginning of testing, the stripper shall be adjusted and tested in accordance with manufacturer's instructions. The use of additional wiring specimens to validate the manufacturer's recommended setting of each wire type is acceptable. Once the adjustments are made, the stripper shall not require further adjustment for the duration of the performance test.

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4.6.1.2 Stripping tool performance. Measure and record the insulated wire or cable diameter at each strip location prior to stripping. The wire or cable diameter or outer configuration shall be in accordance with the wire or cable specification. Strip the specified wire or cable in accordance with the stripper instructions (see 3.7.1). The strip length shall be 0.5 inch, or as specified in the specification sheet. After the wire or cable has been stripped, the stripped insulation shall be carefully removed. Guidance for removing the jacket is given in 6.2. The stripped end shall be viewed under 3X magnification. The stripped cable shall be examined under 3X magnification for wire, conductor, or shield nicks or scrapes, and broken or cut wire/cable conductor or shield strands (see 3.8.1).

4.6.2 Stripping force (see 3.8.2). The stripping force test shall be performed in a standard tensile testing machine with the wire or cable specified in the detail specification sheet being pulled through the stripper. The stripper shall be mounted in a suitable fixture. The wire or cable shall be gripped by the tensile machine jaws and the other end installed in the stripper. Sufficient force shall be applied to cause the wire or cable to be pulled through the stripper. The test machine shall move at a rate of 1 inch/minute. Measure the force to pull 0.5 inch of the wire or cable through the stripper. The stripping force shall not exceed the stripping force specified in the specification sheet.

4.6.3 Shock (see 3.8.3). The stripper with all of the heaviest accessories installed shall be loosely placed in a 12 x 12 x 12-inch box made of 1/4-inch plywood, which has been rigidly fastened to the carriage of a shock test device. The box may be open at the top to facilitate accessibility to the tool. The tool shall be subjected to a shock test in accordance with MIL-STD-202-213, test condition I. Two shocks shall be applied, with the stripper facing the bottom of the box and the other with the stripper rotated 180 degrees. Following the completion of shock testing, insulation strip performance (see 3.8.1) shall be verified by conducting the insulation strip performance test (see 4.6.1).

4.6.4 Humidity (see 3.8.4). The stripper with all accessories required for stripping installed shall be exposed to steady state humidity in accordance with MIL-STD-202-103, condition A. Following the completion of humidity testing, insulation strip performance (see 3.8.1) shall be verified by conducting the insulation strip performance test (see 4.6.1).

4.6.5 Life (see 3.8.5). Only one stripper with applicable conductor stripper die shall be used for the complete life test. The stripper, if an automated testing fixture is used, shall be mounted such that it can be mechanically actuated at a rate of 15 to 20 cycles per minute, where a cycle consists of complete stripping operation of 0.5 inch of wire or cable. Prior to the last strip cycle, examine the wire or cable for stripping performance and the stripper for operation characteristics. The tool may be adjusted every 500 cycles to compensate for blade wear or as needed for inconsistencies in cable dimensions.

4.6.6 Workmanship (see 3.8.6). Examine the stripper and all accessories for sharp edges, burrs and other characteristics that might prevent the stripper performance operation. No parts or accessories that can easily come off through normal handling are permitted.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.3). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The stripper is intended for use in multi-conductor twisted cable, such as those cables in accordance with MIL-DTL-17, SAE AS6070, and NEMA WC27500. The strippers covered by this specification are designed to be used in shock, highly corrosive, and high-temperature environments unique to military applications.

6.2 Jacket removal. To remove the multi-conductor cable jacket material after it has been stripped, bend the cable back and forth 90 degrees to "crack" the jacket where it has been partially cut (scored) (see 3.8.1). Peel the cable back and trim with cable with the appropriate cutting tool. The user may find that the blade depth adjustment for each die is not the same to produce desired results. The use of different blade depth adjustments is acceptable.

6.2.1 Crack and peel. Crack and peel of the cable outer jacket is defined as the manual separation of the cable outer jacket from the multi-conductor cable along the path of cuts (see figures 1 and 2) which were made by the cable stripper die/blade. Initiate crack and peel by bending the cable to complete the separation of the jacket material, and then pull the separated jacket material to the full length of the cuts.

6.3 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Stripper part number (see 1.2).
c. When first article is required (see 3.2).
d. Stripper user instructions (see 3.7.1).
e. Packaging requirements (see 5.1).

6.4 First article inspection process. When first article inspection is required, the equipment used for verification should be initial production units. The contracting officer should include specific instructions in procurement documents requesting verification of first article results. First article inspection report should be re-certified every 3 years.

6.5 U.S. Patent. The strippers specified in this detail specification are covered by the following United States patent. The Government does not have a royalty-free license.

Patent No. U.S. 7,243,429 expiring 17 July 2027.

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6.6 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

6.7 Subject term (key word) listing.

EWIS
Insulation
Tool
Wire

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force - 85

Preparing activity:
Navy-AS

(Project 5120-2019-001)

Review activities:
Army - AV, MI
Navy - EC, OS, SH
Air Force - 11, 20
DLA - GS
Industry association: SAE AE-8C2

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